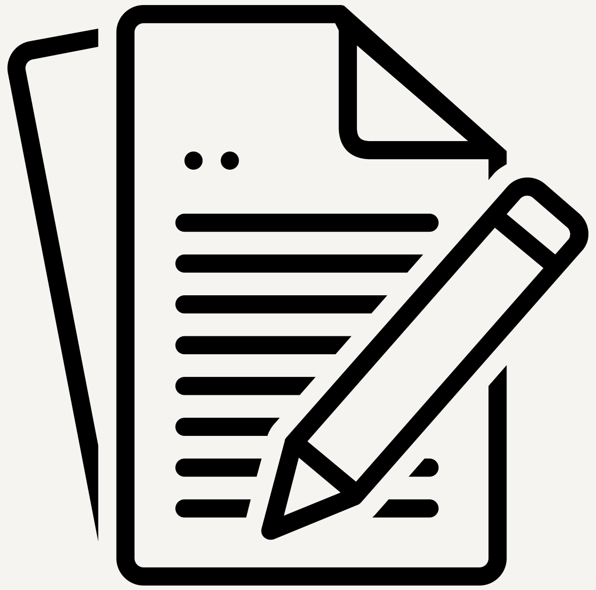
PRESENTED BY:

Trieu Gia Huy

Vo Thanh Nghia

C1: AI Powered Automated Essay Scoring System



PROJECT **SEMINAR**

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PROJECT **OVERVIEW**

OBJECTIVE

- **Goal:** To provide IELTS learners with an AI-powered tool that offers instant, IELTS-style feedback and scoring on writing task 2, helping them improve grammar, coherence, and task relevance to boost their writing band score.
- **Target USER:** IELTS Learners.

PROBLEM

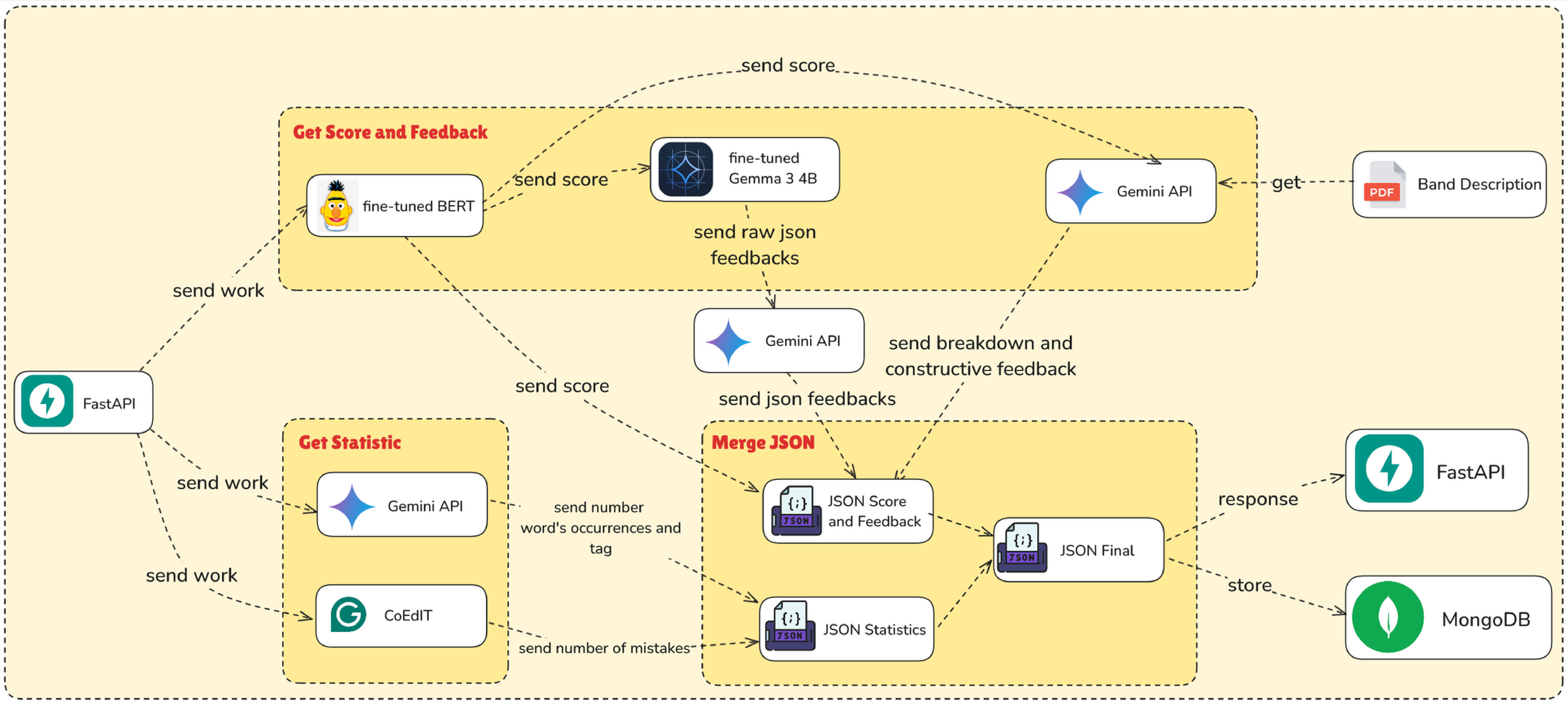
- **Traditional writing practice and feedback methods:**
  - Expensive.
  - Slow.
  - Subjective.
  - Lack of personalized guidance — especially for learners preparing for exams like IELTS.

FEATURE **OVERVIEW**

KEY FEATURES

- **Overall band score (OVR) prediction:**
  - Breakdown into 4 sub-band scores.
- **Analysis of each sub-band following the IELTS writing task 2 format:**
  - Explanation for each band score.
  - General constructive feedback.
- **Statistical insights:**
  - Top repeated content words (to identify lexical repetition and lack of paraphrasing).
  - Number of coherence-related words used.
  - Grammar errors were detected via Grammarly’s pre-trained model.
    - → Suggestions for improvement.
- **Additional features (TBC).**

FEATURE
FLOW



GEMMA 3 REPORT

	Gemini 1.5		Gemini 2.0		Gemma 2			Gemma 3			
	Flash	Pro	Flash	Pro	2B	9B	27B	1B	4B	12B	27B
MMLU-Pro	67.3	75.8	77.6	79.1	15.6	46.8	56.9	14.7	43.6	60.6	67.5
LiveCodeBench	30.7	34.2	34.5	36.0	1.2	10.8	20.4	1.9	12.6	24.6	29.7
Bird-SQL (dev)	45.6	54.4	58.7	59.3	12.2	33.8	46.7	6.4	36.3	47.9	54.4
GPQA Diamond	51.0	59.1	60.1	64.7	24.7	28.8	34.3	19.2	30.8	40.9	42.4
SimpleQA	8.6	24.9	29.9	44.3	2.8	5.3	9.2	2.2	4.0	6.3	10.0
FACTS Grounding	82.9	80.0	84.6	82.8	43.8	62.0	62.4	36.4	70.1	75.8	74.9
Global MMLU-Lite	73.7	80.8	83.4	86.5	41.9	64.8	68.6	34.2	54.5	69.5	75.1
MATH	77.9	86.5	90.9	91.8	27.2	49.4	55.6	48.0	75.6	83.8	89.0
HiddenMath	47.2	52.0	63.5	65.2	1.8	10.4	14.8	15.8	43.0	54.5	60.3
MMMU (val)	62.3	65.9	71.7	72.7	-	-	-	-	48.8	59.6	64.9

(Team el al., 2025).

WHY COEDIT?

	Model	Size	Overall	IteraTeR	Fluency	Clarity	Coherence	Style		
				ITERATER↑	JFLEG↑	ASSET↑	DiscoFuse-Wiki↑	GYAFC(↑/↑)	WNC(↑/↑)	MRPC(↓/↑)
(a)	COPY	-	27.6	29.8	26.7 / 40.5	20.7	30.8	17.6 / 10.6	31.85 / 0	47.4 / 100
	T5-LARGE	770M	24.7	21.1	32.7 / 22.9	35.8	28.01	30.9 / 4.89	13.2 / 0	27.6 / 62.8
(b)	T0*	3B	29.7	26.1	42.2 / 36.1	33.2	32.4	37.9 / 39.3	19.4 / 0	28.3 / 84.1
	Tk-INSTRUCT*	3B	27.3	21.0	35.2 / 26.8	36.9	28.9	35.7 / 43.01	24.2 / 0.1	20.4 / 48.9
	T0++*	11B	32.6	31.5	39.4 / 40.5	33.1	35.5	36.8 / 43.7	21.2 / 0	42.9 / 94.9
(c)	LLAMA	7B	28.2	30.1	27.7 / 3.34	21.8	31.1	18.8 / 89.1	31.9 / 0	5.29 / 64.2
	GPT3	175B	27.4	23.3	38.1 / 2.8	34.8	26.2	36.6 / 87.9	23.4 / 0	0 / 51.7
(d)	ALPACA	7B	28.4	30.4	28.5 / 6.4	22.0	31.1	18.9 / 94.4	31.9 / 0	0 / 77.9
	GPT3-EDIT	175B	41.8	36.1	52.4 / 50.6	32.9	54.0	35.7 / 52.3	50.7 / 17.1	22.6 / 98.7
	INSTRUCTGPT	175B	41.6	32.6	62.4 / 57.2	44.6	47.4	47.8 / 98.2	33.7 / 0.1	16.03 / 98.9
	CHATGPT	-	36.9	28.2	57.6 / 49.4	45.9	40.2	40.7 / 99.6	28.5 / 0.1	**13.4 / 99.0**
(e)	ALPACA (FS)	7B	30.0	30.8	33.03 / 11.3	23.2	33.1	20.6 / 95.4	32.04 / 0	0.1 / 66.7
	GPT3 (FS)	175B	38.4	32.4	50.1 / 4.1	39.2	45.1	43.1 / 97.2	36.7 / 0	0 / 14.5
	INSTRUCTGPT (FS)	175B	45.1	36.2	64.5 / 55.7	**46.3**	55.2	47.3 / 98.8	42.8 / 0	15.9 / 99.5
	CHATGPT (FS)	-	40.1	30.8	58 / 50.6	45.4	51.2	42.3 / 99.6	34.1 / 0	13.3 / 96.1
(f)	ITERATER	570M	31.0	32.8	35.9 / 34.3	21.8	30.1	22.7 / 54.1	34.2 / 0	40.5 / 97.8
	DELITERATER	570M	28.0	29.9	27.5 / 31.2	21.2	32.2	18.1 / 57.8	31.9 / 0	39.1 / 100
	PEER-3B*	3B	41.7	37.1	55.5 / 54.3	30.5	-	-	53.3 / 21.6	-
	PEER-11B*	11B	42.1	**37.8**	55.8 / 54.3	29.5	-	-	54.5 / 22.8	-
(g)	**COEDIT-L**	770M	49.8	35.2	62.4 / 59.3	42.4	75.3	54.6 / 98.0	69.3 / 46.4	23.3 / 99.1
	COEDIT-XL	3B	51.4	36.6	64.5 / 60.7	42.2	**80.5**	**55.1 / 98.3**	70.4 / 48.8	21.3 / 99.6
	COEDIT-XXL	11B	**51.5**	37.1	**65.0 / 61.5**	41.7	78.6	55.1 / 97.2	**71.0 / 51.4**	21.8 / 99.0

(Raheja el al., 2023).

WHY BERT?

Method	Mean Absolute Error (MAE)
Few-shot with specific bands	0.923
Zero-shot	0.687

Bảng 1: Mean Absolute Error (MAE) for 55 new samples across different methods of Sonnet.

Method	Mean Absolute Error (MAE)
Few-shot with specific bands	0.763
Zero-shot	0.768

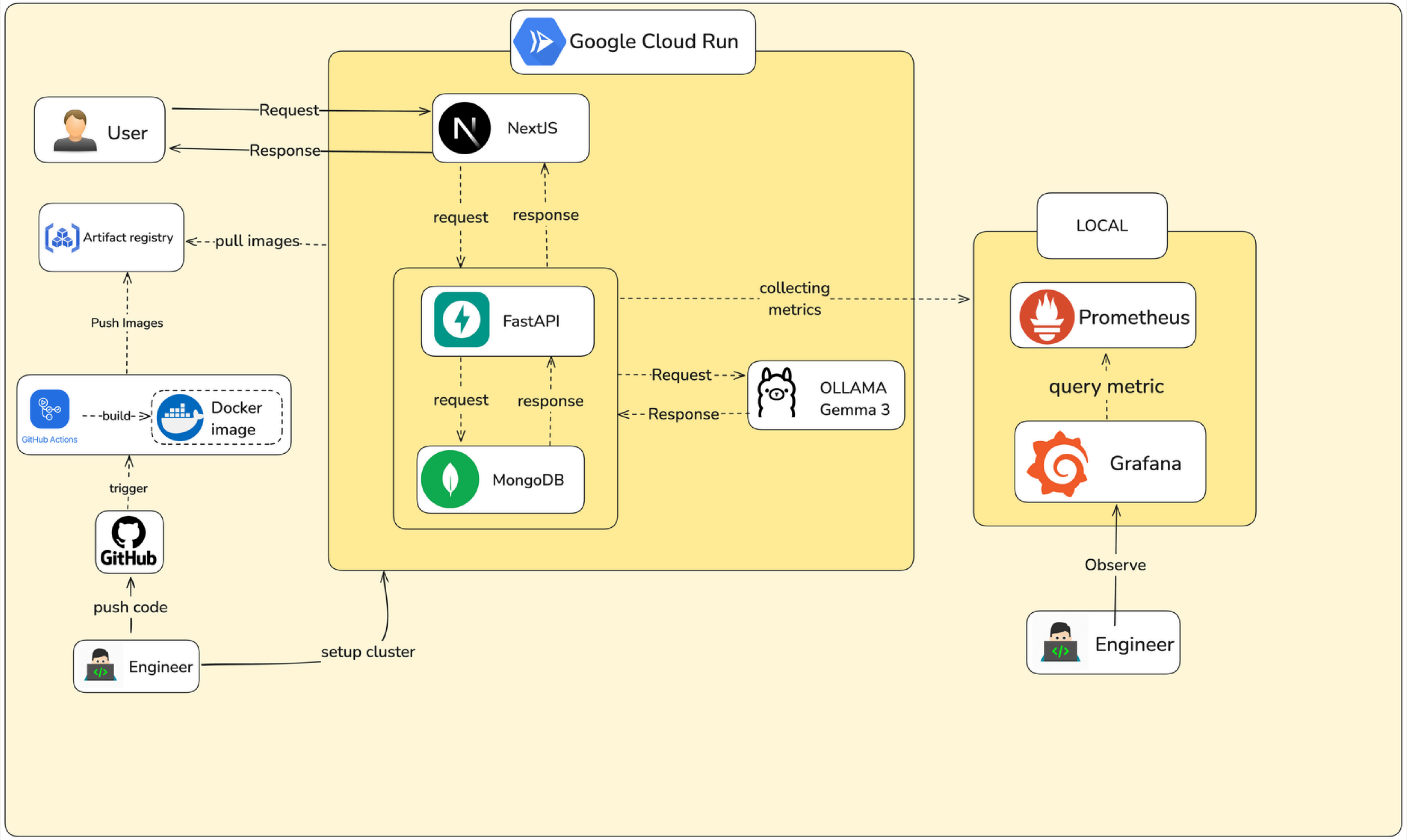
Bảng 2: Mean Absolute Error (MAE) for 55 quality samples across different methods.

Method	Mean Absolute Error (MAE)
Finetuned	0.62

Bảng 3: Mean Absolute Error (MAE) for 55 quality samples using BERT.

(Koraishi, 2024; Xiao el al., 2025).

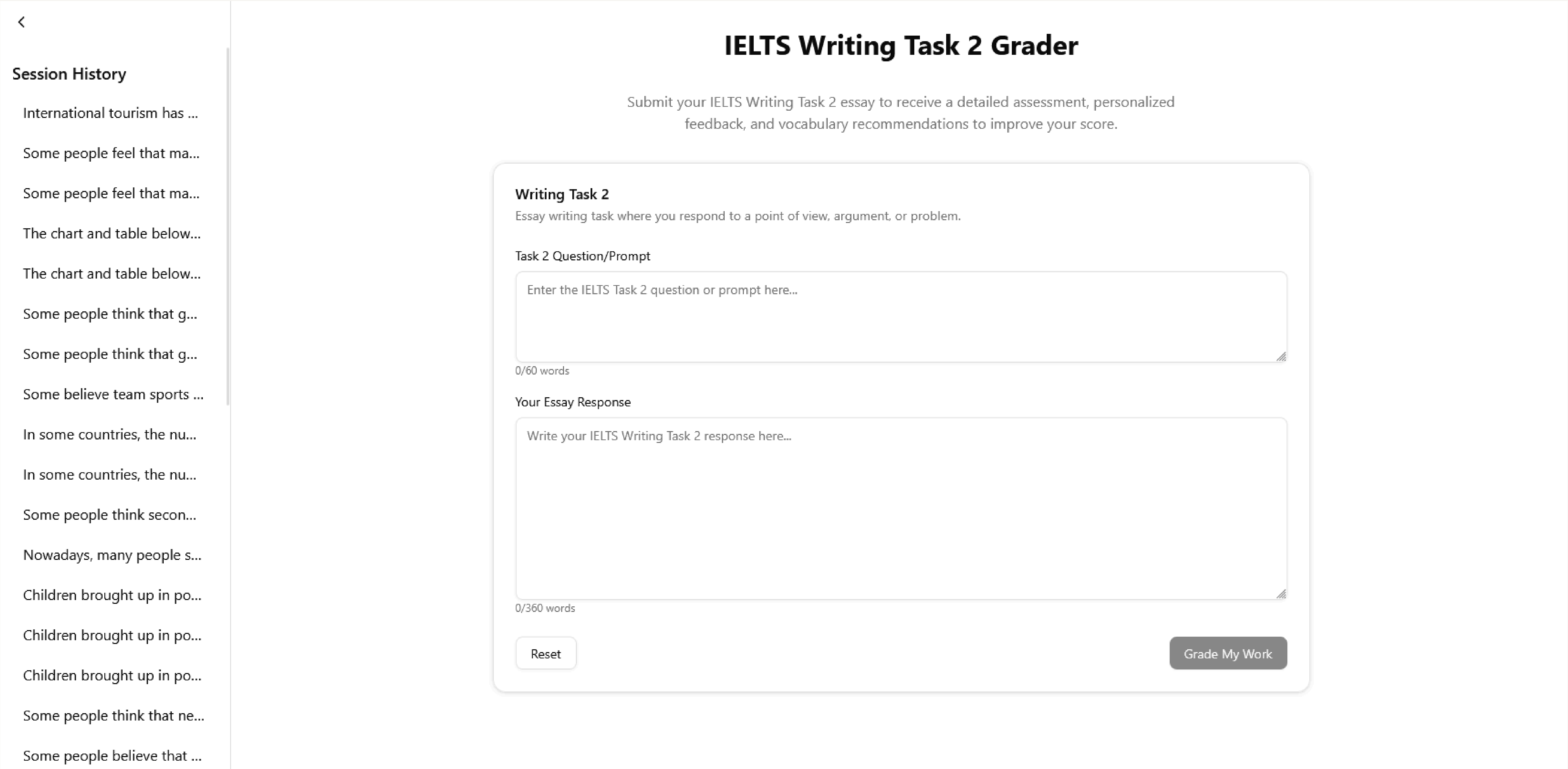
ARCHITECTURE **DIAGRAM**



TECH **STACK**

Components	*Techstack*
Frontend	• NextJS • Tailwind CSS
Backend	• FastAPI
Observation	• Grafana • Prometheus • Loki
Database	• MongoDB
Deployment and containerization	• Docker • Google Cloud run
Model	• Gemini 2.0 Flash • Gemma 3 4B • BERT fine-tuned for scoring • CoEdIT-Large

FRONTEND


Session History
International tourism has ...
Some people feel that ma...
Some people feel that ma...
The chart and table below...
The chart and table below...
Some people think that g...
Some people think that g...
Some believe team sports ...
In some countries, the nu...
In some countries, the nu...
Some people think secon...
Nowadays, many people s...
Children brought up in po...
Children brought up in po...
Children brought up in po...
Some people think that ne...
Some people believe that ...
IELTS Writing Task 2 Grader
Submit your IELTS Writing Task 2 essay to receive a detailed assessment, personalized feedback, and vocabulary recommendations to improve your score.
Writing Task 2
Essay writing task where you respond to a point of view, argument, or problem.
Task 2 Question/Prompt
Enter the IELTS Task 2 question or prompt here...
0/60 words
Your Essay Response
Write your IELTS Writing Task 2 response here...
0/360 words
Reset
Grade My Work

FRONTEND

Your essay with suggestions

International tourism has brought enormous ~~benefit~~ benefits to many places. At the same ~~time~~ time, there is concern about ~~it's~~ its implication on local inhabitants and the environment. Do the disadvantages of international tourism outweight the advantages? In my ~~opinion~~ opinion, the advantages outweight the disadvantages.Firstly, many countries like Egypt or ~~beloud~~ Lebanon live from ~~tourism~~ tourism: lots of people work there as ~~sailmens~~ sailors or tourist guides. These countries ~~without support of tourists wouldn't~~ wouldn be able to function ~~properly.~~ properly without the support of tourists.Secondly, in countries visited by ~~tourists~~ tourists, there are plenty of places where people just ~~can't~~ can pass because of mere animals or plants. Another thing is that people like traveling and seeing new exotic places. They like ~~we~~ us on the beach or ~~swim~~ swimming in ocean.Furthermore, tourism is now ~~more~~ a growing industry ~~higheing~~ employing thousands of people. There are making new places to work and to have fun. But on the other hand, people often forget that they aren't the only things on the planet. Many tourists are living garbage just anywhere.Some of them wan't an exotic summer so they pay for illegal things like dedd animals or some sculpture. To sum up I think international traveling is a good thing but people must realise that these there are something else besides them.They need to know that flora and fauna needs to be protected. People have to enjoy their holidays but alsow protect environment.

FRONTEND

Your IELTS Score

OVERALL BAND
6.0

TASK ACHIEVEMENT	COHERENCE & COHESION	LEXICAL RESOURCE	GRAMMATICAL RANGE
6.0	6.0	6.0	6.0

Task Achievement

Score: 6.0

Strengths

- Addresses all parts of the task, presenting both advantages and disadvantages of tourism.
- Presents a clear position ('advantages outweight the disadvantages').

Evaluation

- The essay addresses the prompt and presents a clear opinion, stating that advantages outweigh disadvantages
- The essay provides some supporting arguments, such as employment opportunities and the enjoyment of travel
- However the arguments are somewhat underdeveloped and lack specific examples or elaboration
- The connection between the arguments and the overall opinion could be stronger

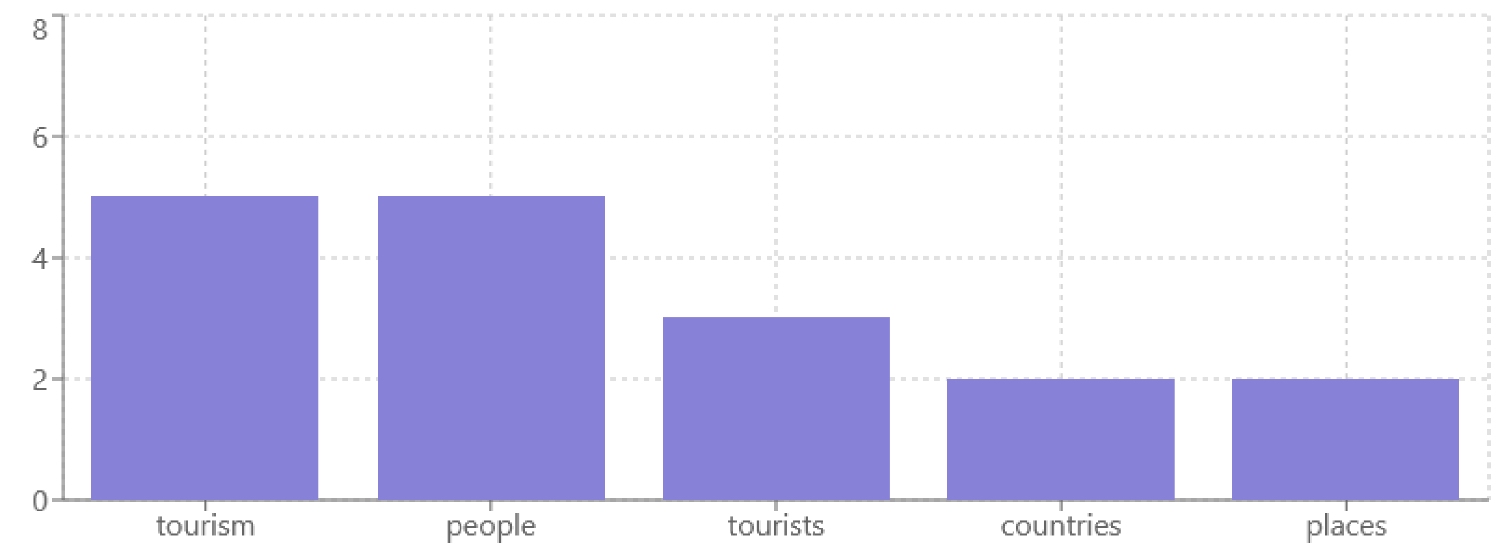
Areas for Improvement

- Some ideas are underdeveloped and lack specific examples (e.g., 'countries visited by tourists are plenty of places where people just can't pass because of mere animals or plants').
- The supporting arguments for the advantages of tourism could be more thoroughly explained and extended.
- The discussion of disadvantages is somewhat superficial and lacks depth.
- The conclusion could be stronger by providing a more nuanced summary of the main points.

Word Usage Analysis

COHERENCE WORD COUNT
2

Top Repeated Content Words



tourism 5 times	people 5 times
tourists 3 times	countries 2 times
places 2 times	

CHALLENGES & **SOLUTIONS**

CHALLENGES

- Dataset, especially for constructive feedback and explaination.
- Relevance between question and essay.
- Gemini API is not stable

SOLUTIONS

.

FUTURE **WORK**

- Explore more scoring models.
- Integrate real-time chatbot assistance.
- Discover more intelligent word suggestion system (with more than one word).
- Gamified writing experience.
- Support teacher role, with teacher dashboard & analytics.
- User report section.

Reference

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Thank You!

Do you have any questions for me before we go?